B TAJ DPM Tutorial

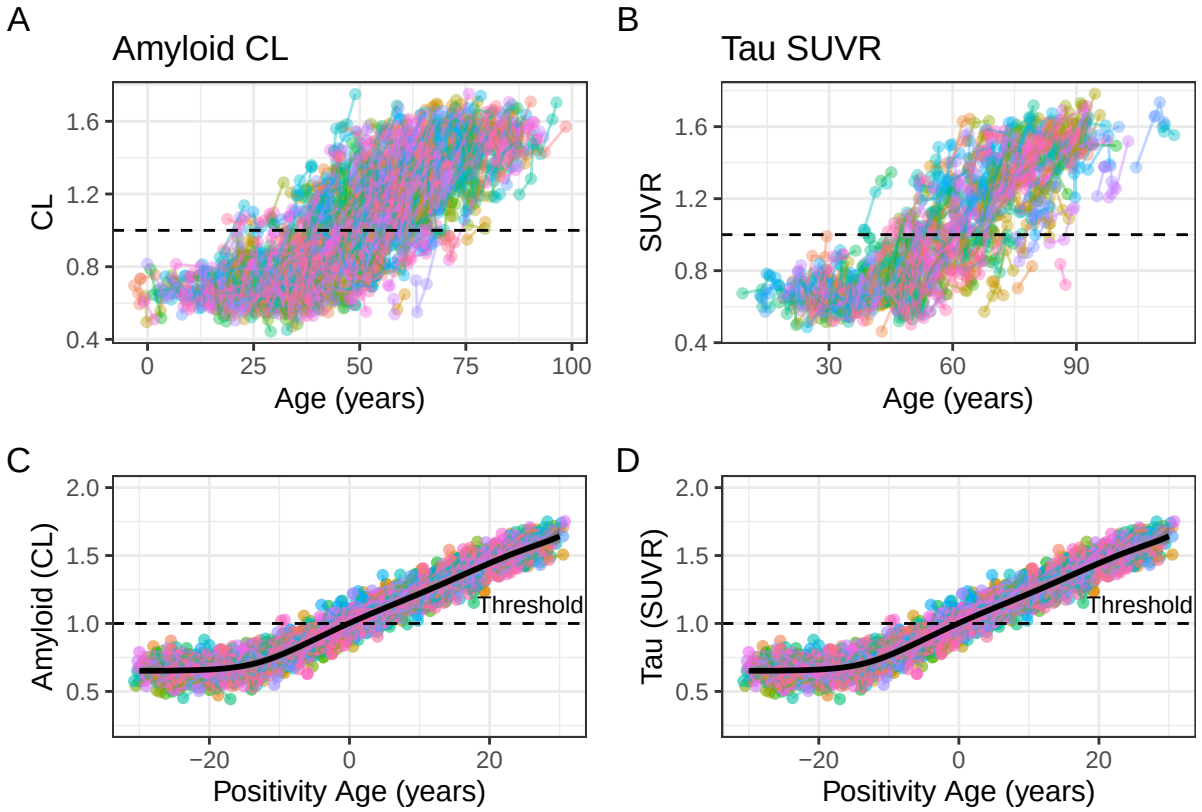
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2026-06-01

Simulation

We simulate longitudinal amyloid and tau biomarker data for 1,000 subjects and fit the Bayesian disease progression model using MCMC. The subsequent plots compare trajectories on the calendar-age scale with posterior-estimated disease-time trajectories after alignment, including uncertainty bands.

```
set.seed(123)
res <- simu(varying_coef_code_sim,
            varying_coef_code,
            n_subjects = 1000,
            sa = 10,
            sy = 0.07,
            outer_iter = 1,
            niter = 2000, nburnin = 1000, #thin = 1,
            nchains = 1,
            save_res = F)
```



Posterior prediction

We generate a separate simulated cohort and use posterior draws from the fitted model to predict disease progression for new subjects. The forward-prediction analysis conditions on amyloid observations below the specified threshold, estimates individual time shifts, and visualizes predicted trajectories with posterior uncertainty alongside observed measurements.

```
source('posterior code/real data bivariate plus survival data cv.R')

dta_amy <- data.frame(CENTILOIDS = simudata$cModelsim$y_sparse_amy, RID_num = data_1$constants$subject_a
                      age_at_scan = simudata$cModelsim$t_amy, RID = simudata$constants$subject_amy)

dta_tau <- data.frame(SUVR = simudata$cModelsim$y_sparse_tau, RID_num = data_1$constants$subject_tau,
                      age_at_scan = simudata$cModelsim$t_tau, RID = simudata$constants$subject_tau)

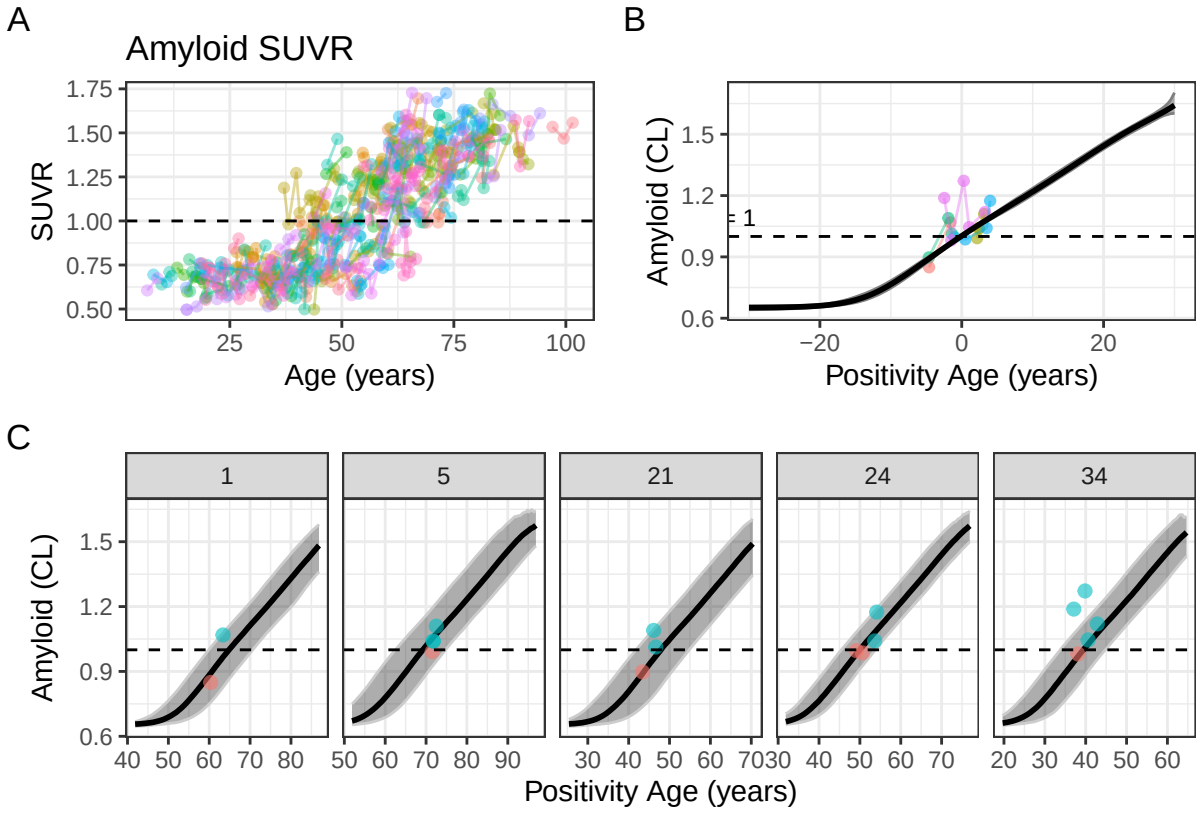
# Match the observation column names read by realdata_pred().
dta_amy$SUVR <- dta_amy$CENTILOIDS
dta_tau$mesial_temporal_tau_pet_metaroi <- dta_tau$SUVR
dta_surv_df <- data_surv
dta_surv_df['RID'] <- dta_surv_df['subject_id']
dta_surv_df['time'] <- dta_surv_df['t']
pred_listf <- realdata_pred(X_amy = simudata$constants$Xsub_amy,
                            X_tau = simudata$constants$Xsub_tau,
                            1:50, simudata$constants$tau_amy_link, dta_surv_df, res$samples, simudata,
```

```r
# Use the observation ages from the selected prediction data.
pred_listf$simudata_cv$constants$t_amy <- pred_listf$simudata_pred$constants$t_amy
pred_listf$simudata_cv$constants$t_tau <- pred_listf$simudata_pred$constants$t_tau

# Map each tau observation to the corresponding subject in the joint alpha matrix.
pred_listf$simudata_cv$constants$subject_tau <-
  pred_listf$simudata_cv$constants$tau_amy_link[
    pred_listf$simudata_pred$constants$subject_tau
  ]

pred_listf$simudata_cv$inits$gamma_amy <- NULL
pred_listf$simudata_cv$inits$gamma_tau <- NULL


source('posterior code/real data mcmc_posterior.R')
source('posterior code/posterior predictive distribution.R')
posterior_res_forward <- mcchains(posterior_predictions_code,
                                  pred_listf$simudata_cv,
                                  monitors = c("alpha", "gamma_amy_post", "gamma_tau_post",
                                               "f_pred_amy_post", "f_pred_tau_post",
                                               "rho_post"),
                                  niter = 1000,
                                  nburnin = 500,
                                  thin = 1,
                                  nchains = 1,
                                  fullres = TRUE)
```


A
Amyloid SUVR
SUVR
1.75
1.50
1.25
1.00
0.75
0.50
25
50
75
100
Age (years)
B
Amyloid (CL)
1.5
1.2
1
0.9
0.6
−20
0
20
Positivity Age (years)
C
1
5
21
24
34
Amyloid (CL)
1.5
1.2
0.9
0.6
40 50 60 70 80
50 60 70 80 90
30 40 50 60 70
30 40 50 60 70
20 30 40 50 60
Positivity Age (years)